



Product Development of a Substrate Mixer and Portioning Device

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Bachelor's thesis
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Substrate for mushroom growing

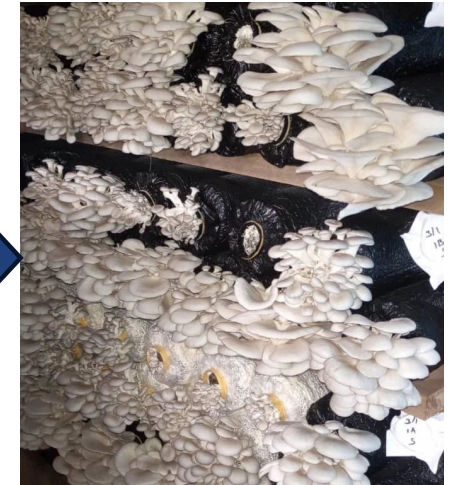
Mix and water agricultural waste products



Bag, pasteurize and insert mushroom spawn



Harvest edible mushrooms



Why should mushrooms be cultivated in developing countries?

Circular Economy: "From waste to money"

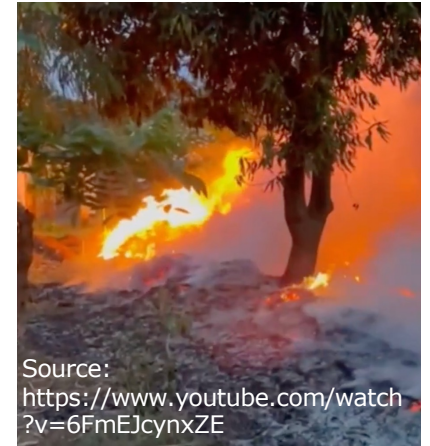
Agricultural waste



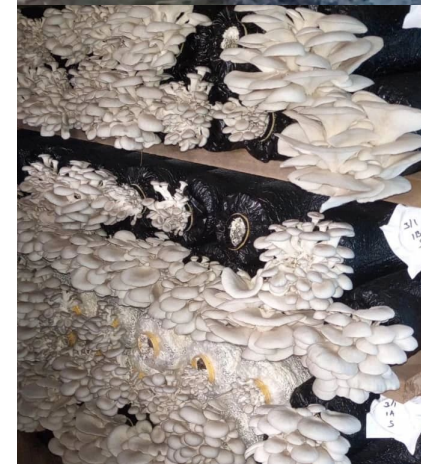
Waste disposal
Environmental pollution

Mushroom growing

Income generating



Source:
<https://www.youtube.com/watch?v=6FmEJcynxZE>



How should mushrooms be cultivated in developing countries?

Ugandan baseline



Manual substrate preparation:

- High labor cost
- Inconsistent yields

Mechanized substrate preparation:

- High capital requirement
- High energy consumption

Industrial processing



How should mushrooms be cultivated in developing countries?

Ugandan baseline



Constraints in Kampala:

- Low capital (smallholders)
- Limited material availability
- Unreliable energy supply
- High energy cost

Goal of this project

Industrial processing



Requirements for the device (Research Questions)

Mixing

How can heterogeneous agricultural residues be processed to achieve a homogeneous substrate composition throughout the mixing vessel?

Watering

How can the substrate be uniformly hydrated to achieve a target moisture content of 60-70 wt%?

Portioning

How can the processed substrate be portioned into bags with deviations within a tolerance of ± 10 wt%?

Product Development - Methodology

“Simplicity is the Reproducibility.” – Leonardo da Vinci

Globally available materials



Simple manufacturing



Low-tech & low-cost



Product Development

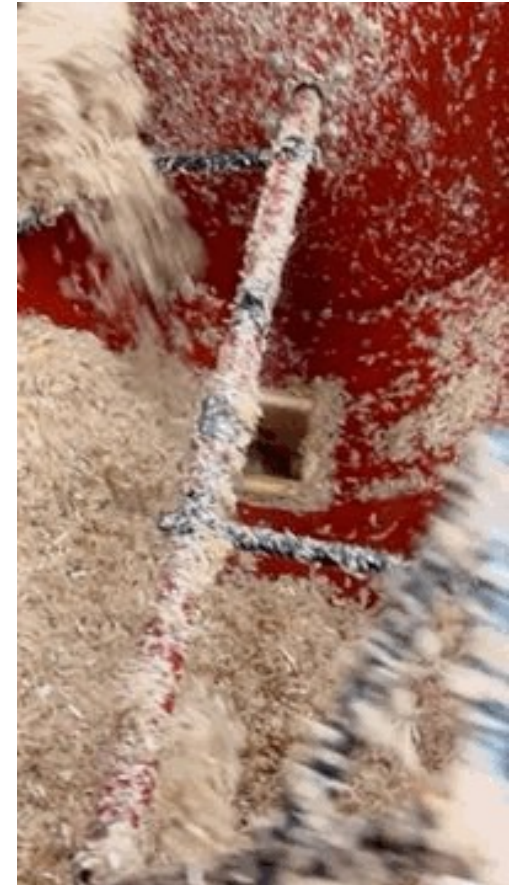
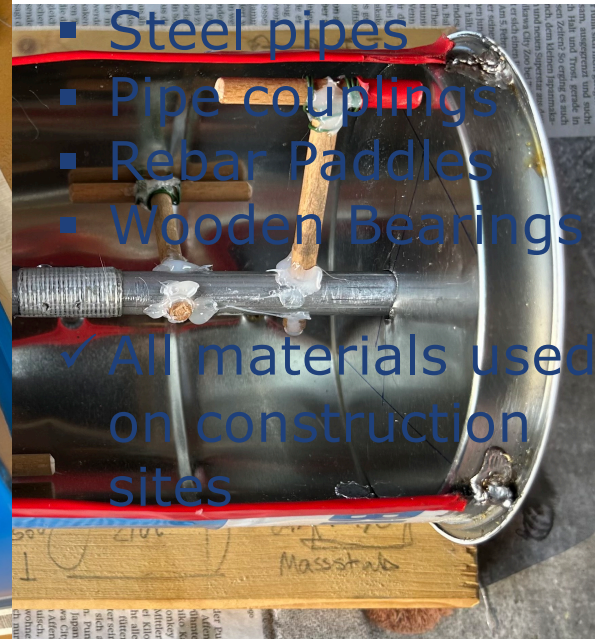
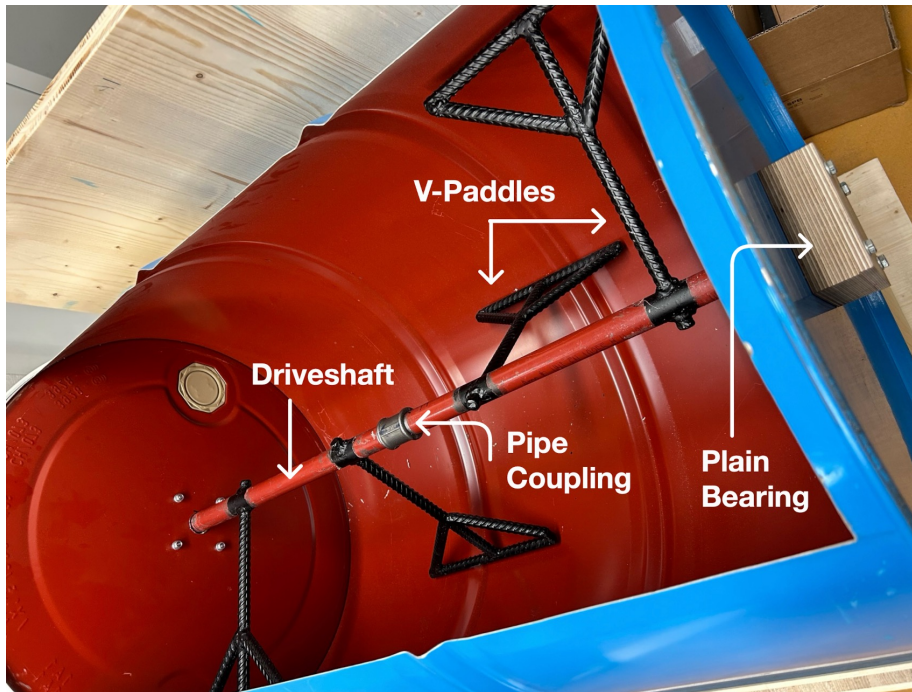


- Oil drum
- Wooden panels
- ✓ Globally standardized materials & items

Product Development - Mixing

Final agitator

Model scale 1:5 – Testing different paddle geometries



Product Development - Watering

- Hydrostatic Reservoir
- Manifold (sprinkler)
- ✓ Gravity driven
(no energy supply)



Product Development - Portioning

Sealing Valve



Conveyance Duct



Slide Valve

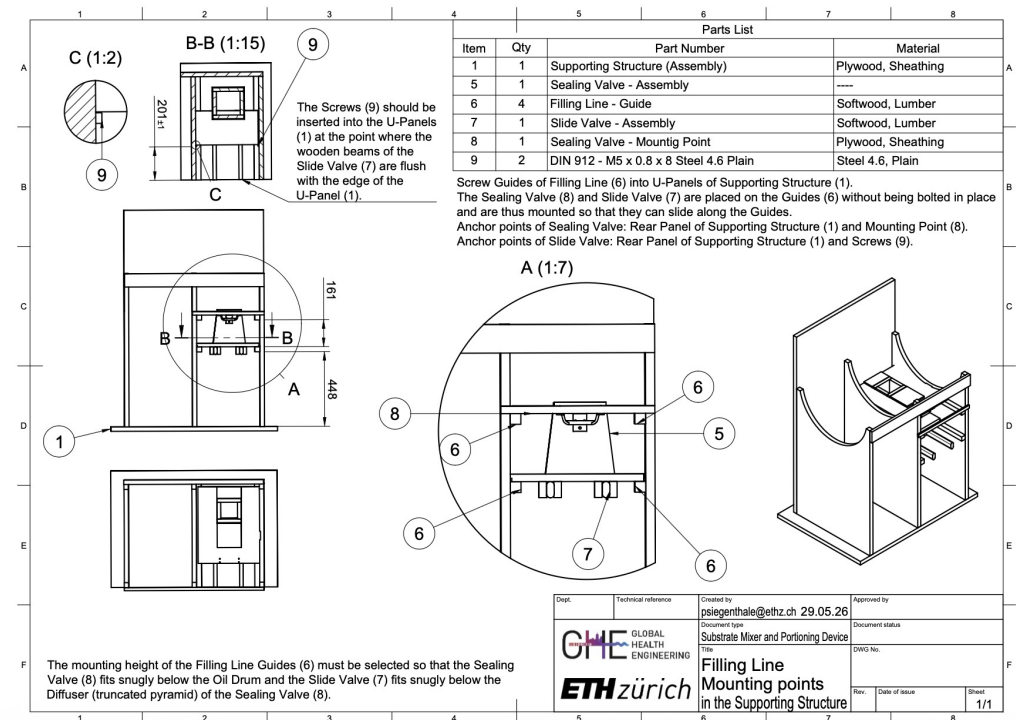
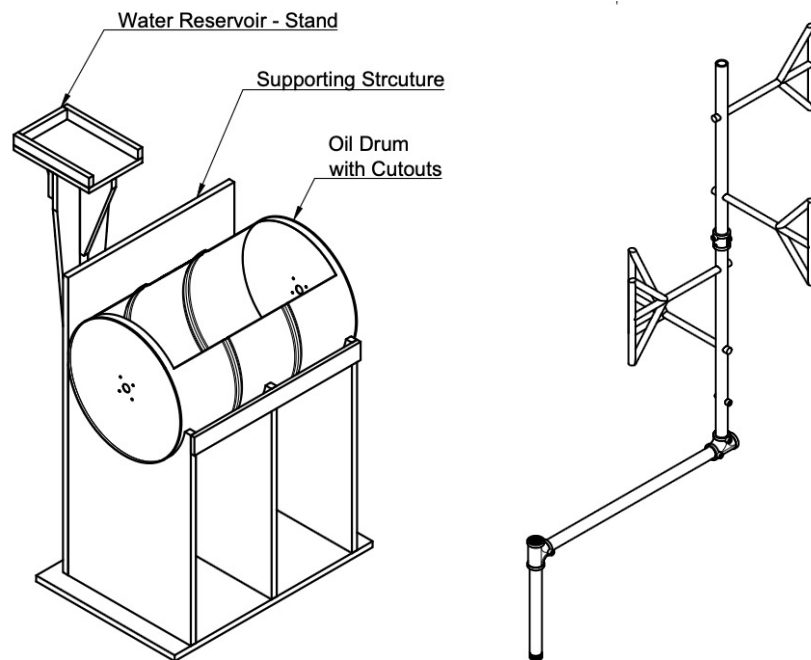


Product Development - Portioning

Substrate blockages in conveyance duct

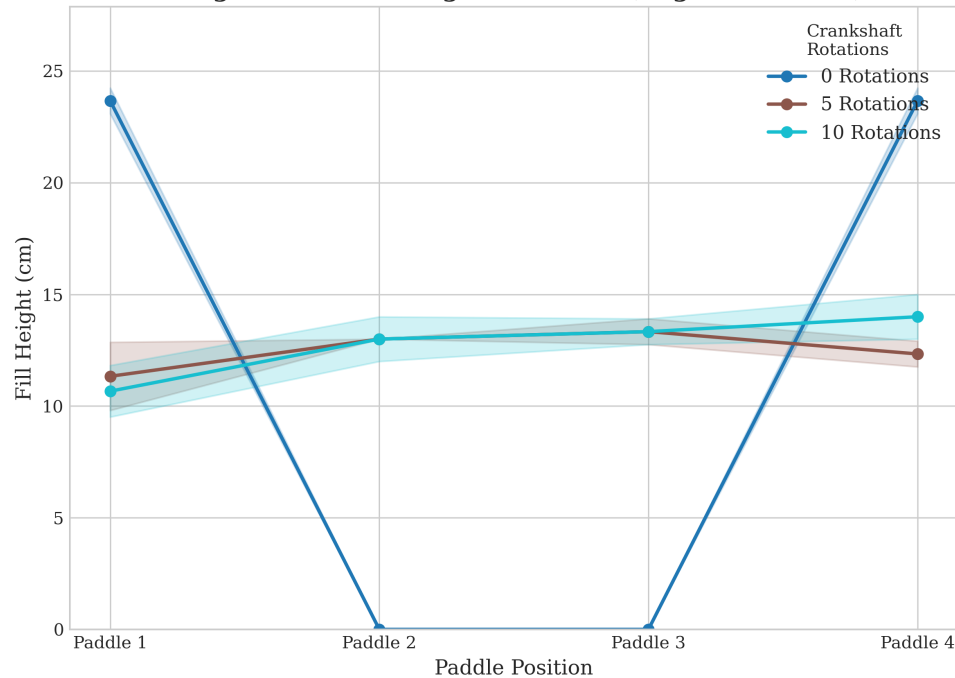


Deliverables – Technical Drawings, User Manual

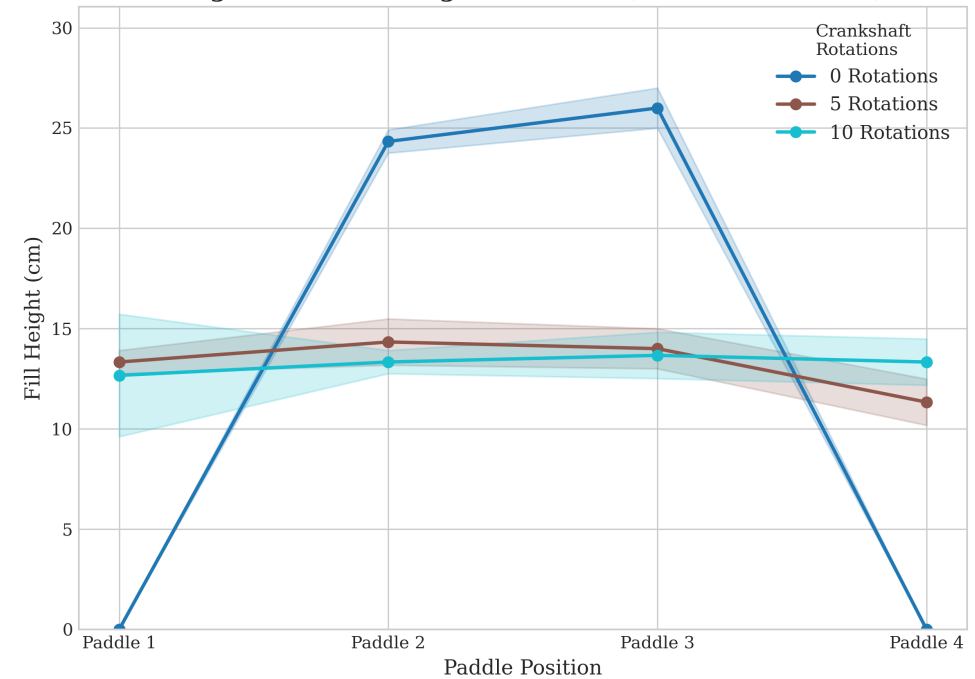


Mixing – Longitudinal Transport

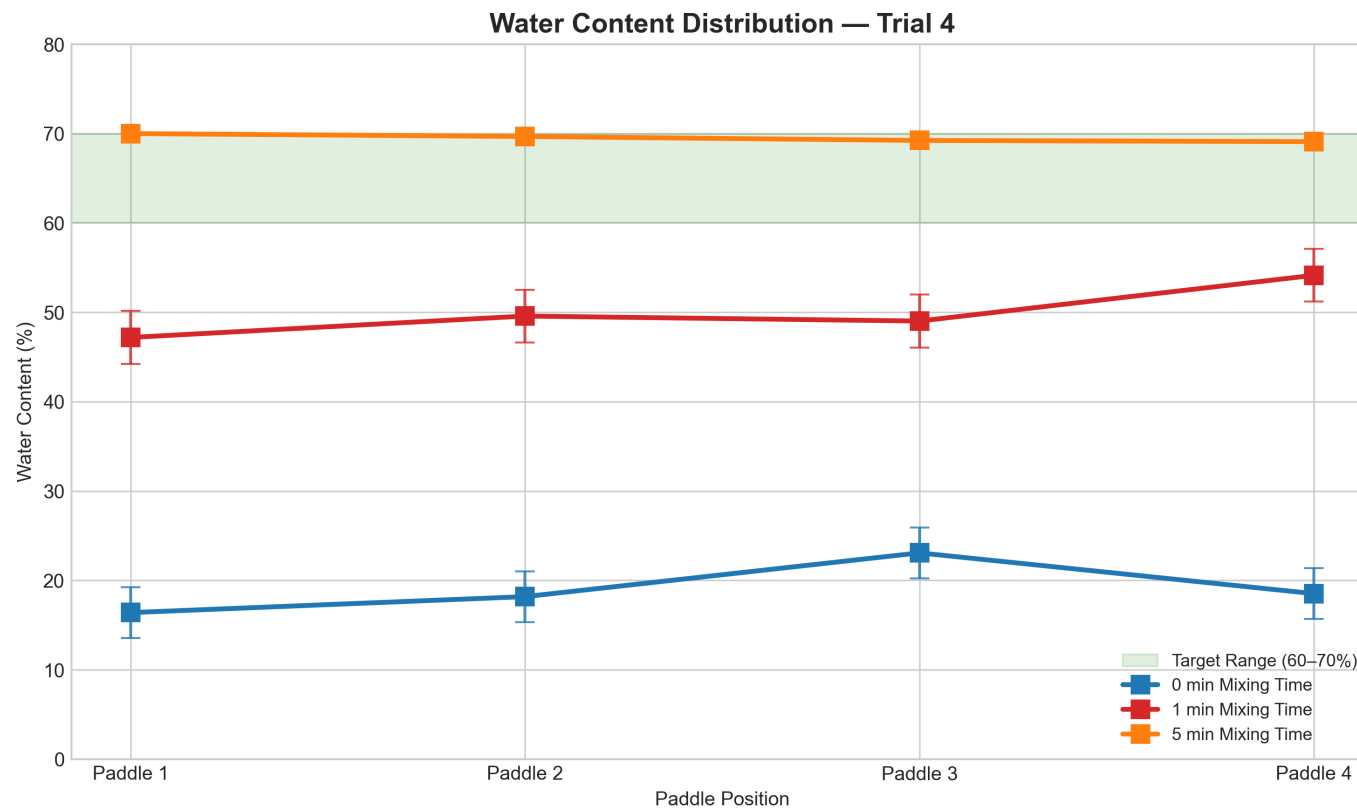
Longitudinal Fill Height - Test 1.2 (Edge Placement)



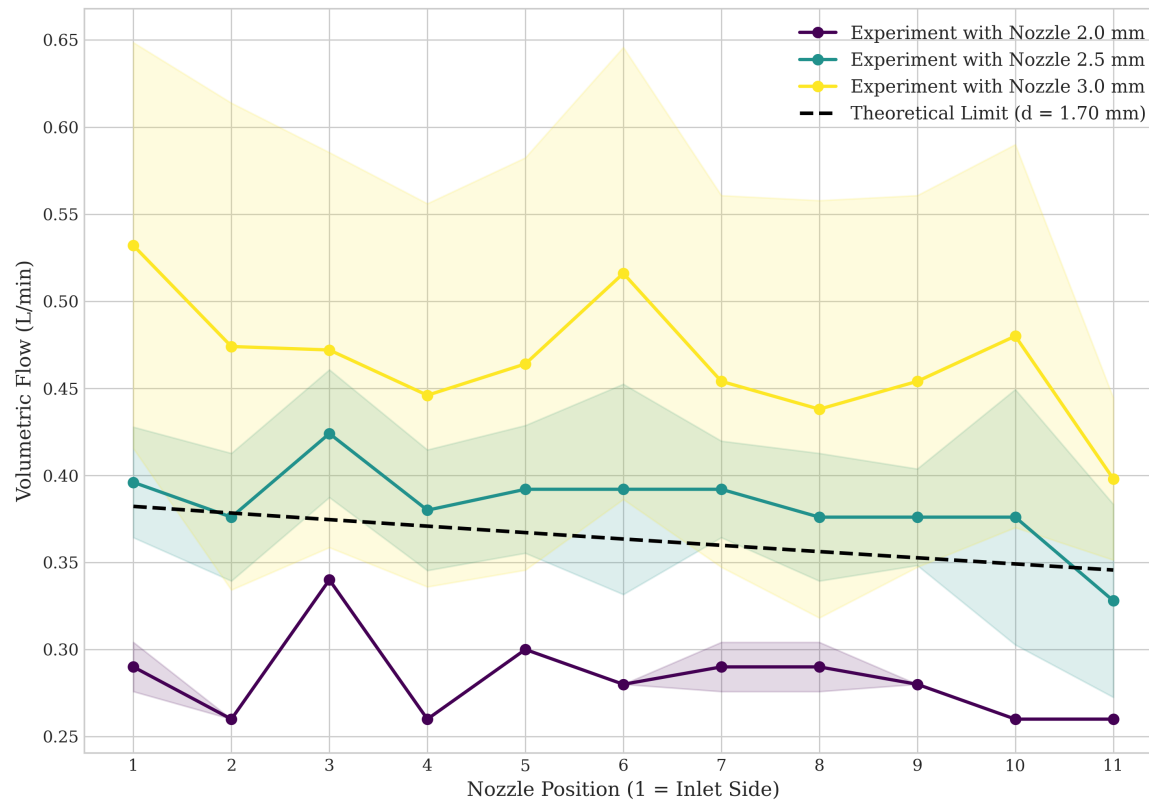
Longitudinal Fill Height - Test 1.1 (Center Placement)



Mixing – Water Content Distribution



Watering – Nozzle Outflow



Portioning – Bagging Performance

Trial	Prototype	Mean (kg)	SD (kg)	RSD (%)	Loss (wt%)	Note
1	PE-Pipe	0.414	0.153	36.89	45.42	all bags
2	Diffuser	1.367	0.105	7.69	8.74	all bags
3	Diffuser	1.146	0.186	16.25	3.24	all bags
3	Diffuser	1.235	0.105	8.52	—	bags 6–18

Conclusion – Working Prototype

Hand-driven, low-cost mechanical solutions can successfully standardize substrate processing.

Manual Baseline



Mechanized & Standardized Solution



Big reduction of labor estimated

Conclusion – Fulfilled Research Questions

- **Homogeneous Mixing and Watering:**
 - ✓ Water content deviations (0.45 ± 0.06) wt%.
- **Consistent portion sizes:**
 - ✓ Mass deviations below ± 10 wt% threshold.

Outlook – Recommendations

- **Field Validation in Uganda:**

- Analysis of user acceptance, reproducibility, durability.

- **Further Testing and Improving:**

- Behaviour of different substrate ingredients.
- Investigating need for agitator gearbox or increased stability.

ETH zürich

Final Prototype

